

CH5440

Description

Use of data analysis for process performance enhancement Mathematical background: Sources and characterization of errors in data, Random variables, probability density functions, estimation, confidence intervals and hypothesis testing, stochastic signals, frequency domain analysis of signals, measures of nongaussianity. Process Modeling: Model structures, linear regression, Nonlinear regression, principal Component Analysis, Independent Component Analysis. Applications: Parameter estimation in linear and nonlinear processes, Data Reconciliation, Continuous/Batch process monitoring using MSPC, controller performance monitoring, fault diagnosis, chemometrics, biomedical and speech signal processing.

Course Content

TextBooks

1. Johnson R A and D W Wichern, Applied Multivariate Statistical Analysis, Prentice Hall, 2002.
2. Montgomery, D C and G C Runger, Applied Statistics and Probability for engineers, Wiley, 2003

Reference Books

NIL

Prerequisite

NIL